

Decision tree learning

Aim: find a small tree consistent with the training examples

Idea: (recursively) choose “most significant” attribute as root of (sub)tree

```
function DTL(examples, attributes, default) returns a decision tree
  if examples is empty then return default
  else if all examples have the same classification then return the classification
  else if attributes is empty then return MODE(examples)
  else
    best ← CHOOSE-ATTRIBUTE(attributes, examples)
    tree ← a new decision tree with root test best
    for each value  $v_i$  of best do
      examplesi ← {elements of examples with best =  $v_i$ }
      subtree ← DTL(examplesi, attributes – best, MODE(examples))
      add a branch to tree with label  $v_i$  and subtree subtree
  return tree
```